

Abstract

Task-aware camera control can improve downstream vision performance, but learning suitable policies requires data covering many camera configurations, which are costly to capture and difficult to reproduce realistically through image recapture. We introduce a physically grounded simulating pipeline that models exposure-dependent effects such as clipping, sensor noise, and quantization from HDR images, opening the opportunity to extend existing datasets with HDR data, such as Cityscapes or PASCALRAW, for camera parameter optimization.

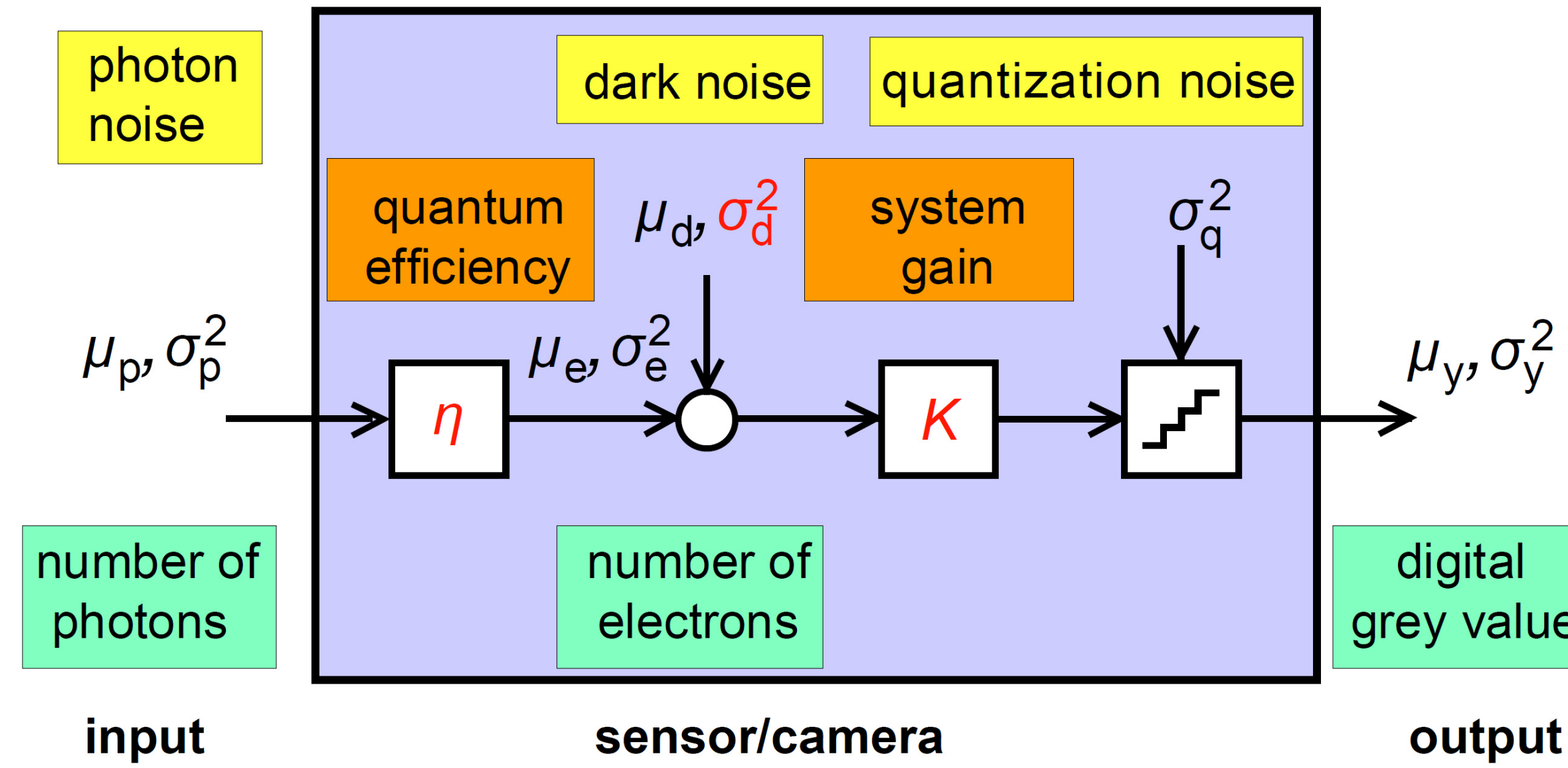
The Problem Setting

- End-to-end parameter optimization in visual pipelines as open task
- Data with different camera parameters and task annotations scarce
- ImageNet-ES Diverse containing photographs of printed out images, but not accurately representing dynamic range changes for different camera parameters [1].
- Current sensor simulation implementations are only available in Matlab [2], and have not been used for parameter optimization.

Our Solution: Simulating varying exposure triangle parameters utilizing HDR data, simulated or captured, creating data for camera parameter optimization from existing HDR datasets.

EMVA Standard 1288 as Starting Point

Our simulation pipeline follows the EMVA 1288 linear camera model [3], modeling the conversion from incident photons to digital pixel values, including sensor characteristics and noise.



Parameters used in our Camera Simulation

Category	Parameter	Value(s)
Exposure Triangle	ISO	$k_{ISO} = ISO/100, \quad ISO \in \{250, 2000, 16000\}$
	Exposure time [s]	$t_{exp} \in \{1/4, 1/60, 1/1000\}$
	Aperture [f]	$N \in \{5.0, 9.0, 16.0\}$
Sensor Properties	System gain	$k_{Gain} = 0.425$
	Dark noise sigma	$\sigma_d = 2.43$
	Saturation capacity	$c_{max} = 9091$
	Quantum efficiency	$\eta = 50\%$
Calibration	HDR-to-radiance scaling factor	α

Our Simulation Pipeline

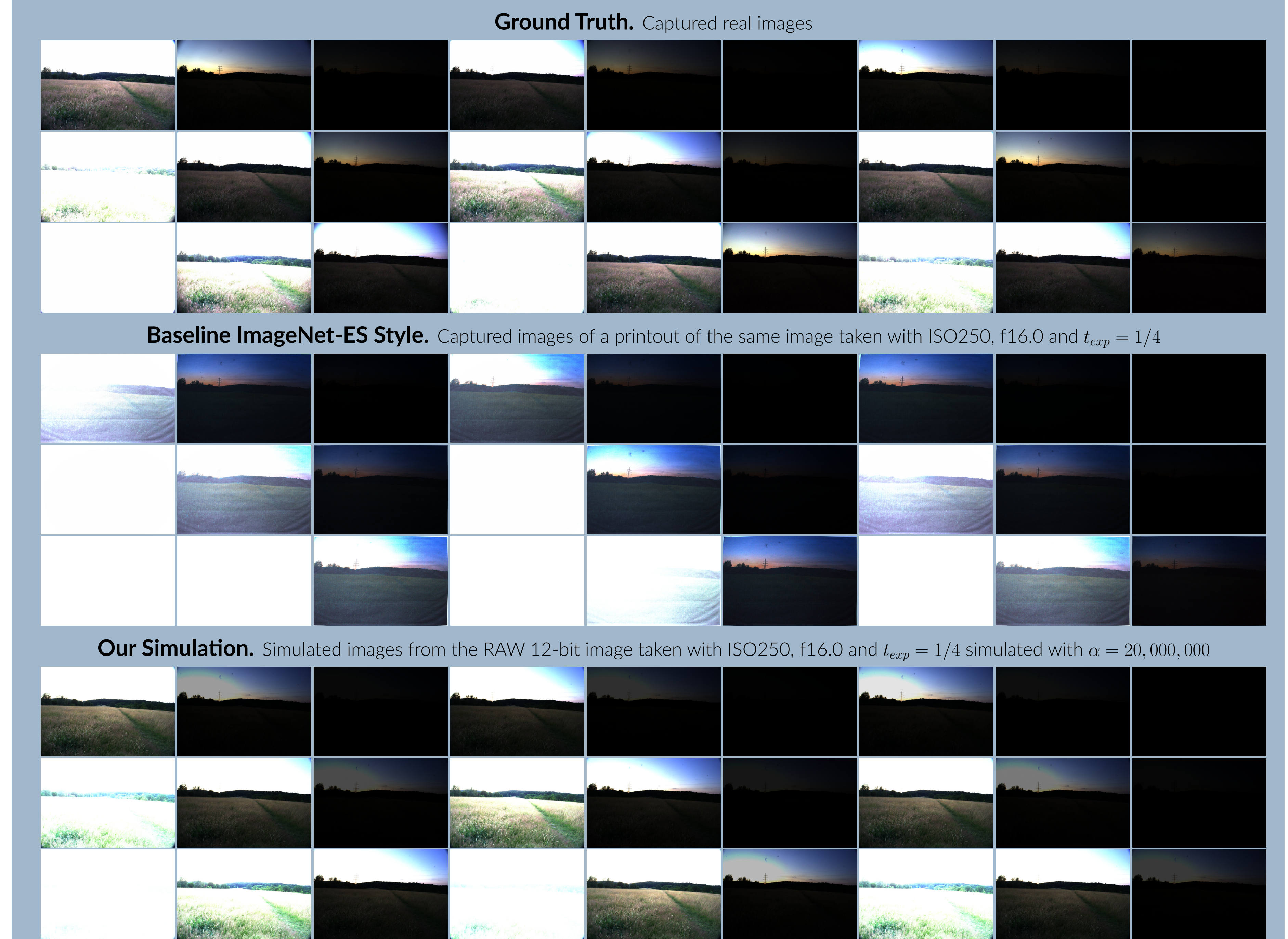
- Irradiance.** Using the thin lens approximation and the discussed calibration factor α we obtain an irradiance on the sensor plane via
$$E(x, y) = \frac{\pi}{4N^2} \cdot \alpha I_{HDR}(x, y). \quad (1)$$
- Photoelectrons.** The expected number of generated photoelectrons becomes
$$\mu_e(x, y) = E(x, y) t_{exp} \eta. \quad (2)$$
- Noise.** Photon shot noise is modeled as Poisson distributed, while read noise is modeled as Gaussian. Thus, for
$$n_{ph}(x, y) \sim \text{Poisson}(\mu_e(x, y)), \quad n_r(x, y) \sim \mathcal{N}(0, \sigma_d^2), \quad (3)$$
we assume a camera signal given by
$$s(x, y) = k_{ISO} k_{Gain} n_{ph}(x, y) + n_r(x, y). \quad (4)$$
- Quantization.** Finally, the analog sensor response is converted into an 8-bit image through a linear conversion factor k , saturation, and linear quantization Q_{8bit} :
$$I_{sim}(x, y) = Q_{8bit}(\text{clip}(s(x, y), 0, c_{max})). \quad (5)$$

HDR Images as Radiance Approximation

As input to our pipeline any HDR images, captured or synthetic, can be used. We present simulations with images we took ourselves, for which we also obtained a ground truth to compare against, the upcoming RDR2 synthetic dataset, S2R-HDR [4], and PASCALRAW [5].

We assume that the linear HDR image I_{HDR} represents the scene radiance up to a calibration factor α . For calibration, we compare simulated images with photographs captured using the Sony $\alpha 6000$ under visually similar illumination choosing α accordingly.

Comparison: Real Captures, Printed and Recaptured, and Simulated Images

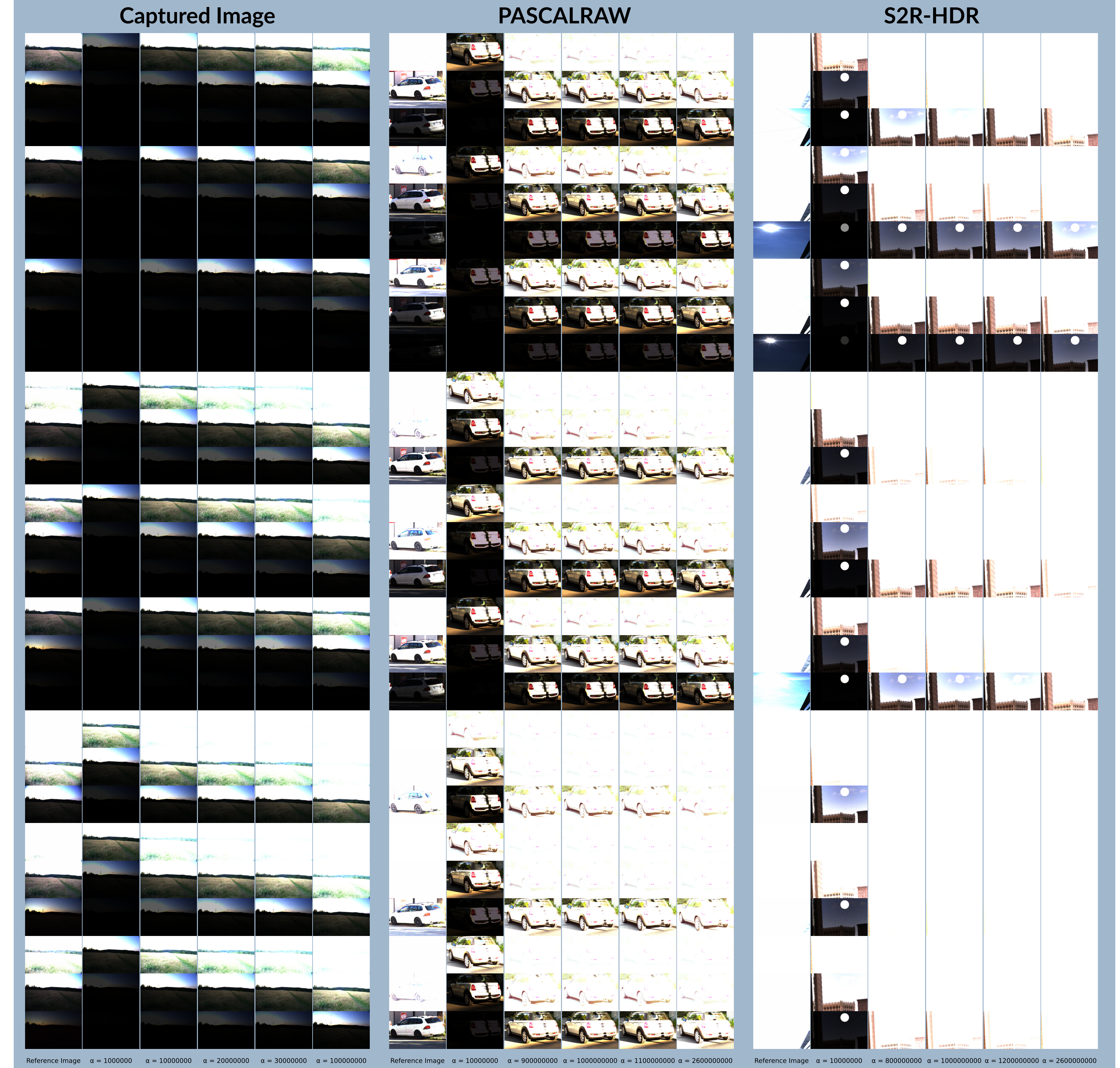


Simulation Results on Synthetic and Real Datasets



*Rows correspond to ISO values (top to bottom: 250, 2000, 16000). Columns are arranged by aperture groups (left to right: f/5.0, f/9.0, f/16.0), with shutter speeds within each group ordered as 1/4 s, 1/60 s, and 1/1000 s (left to right).

Calibrating α



*Columns correspond to different alpha values tested during calibration, the reference image to calibrate to being shown in the first column. The 4th column contains the final images with the calibrated α value. For PASCALRAW and S2R-HDR we cropped the images to a shared motive for calibration.

What Comes Next

Optimizing Calibration & Different Camera Models

- Investigate HDR-Irradiance relation more deeply
- Capture more ground truth with different camera models
- Find calibration parameter relation to sensor properties of capturing camera for real datasets
- Optimize calibration process

Verifying and Extending the Model

- Include fixed-pattern noise
- Validate accuracy of simulated noise
- Include depth-of-field effects with depth maps
- Include motion-blur from flow fields

Extend Existing Datasets for Task-aware Camera Parameter Optimization

- Cityscapes [6] (contains depth maps)
- S2R-HDR [4] (contains depth maps and flow fields)
- RDR2 (contains depth maps)

TL;DR

We are developing a **physically grounded camera simulation pipeline** based on the EMVA 1288 standard for generating images as captured with **variable camera parameters from HDR scene representations**, already showing promising results on **real and synthetic scene data**.

References

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